

1. System Architecture & Multi-Tenant Role Isolation

Role Key	Portal Access	Data Scope	Authoritative Auth Source	Default State
CITIZEN	/dashboard, /report	Own complaints & public geo-feed	Firestore UID -> Supabase users	Active (Default)
DEPARTMENT	/department/dashboard	Department-scoped issues only	Supabase users.department_id	Assigned by Admin
ENGINEER	/engineer/dashboard	Assigned work orders & dept scope	Supabase users (role=ENGINEER)	Assigned by Admin
ADMIN	/admin/dashboard	Global unconstrained system oversight	Supabase users (role=ADMIN)	Elevated Root Authority

All civic issues traverse an immutable 17-state finite state machine with strict transition authorization enforced at both the API layer and the PostgreSQL database trigger layer:

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[IN_PROGRESS]

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Role claims embedded in client JWTs or request payloads are strictly ignored. All authorization checks query the PostgreSQL users table via the secure backend service-role API gateway (functions/authMiddleware.js).

CIVICCONNECT SYSTEM MODULE 01

Citizen Engagement & Reporting Module

ROLE DEFINITION: CITIZEN

1.1 Module Objective & Functional Scope

The Citizen Module empowers residents to register municipal grievances, capture high-resolution photographic evidence with embedded geolocation, receive real-time updates via WebSockets, and verify or reopen resolved complaints through an automated feedback loop.

1.2 End-to-End Citizen Lifecycle Stages

Stage 1: Identity Creation & Auto-Provisioning

- **Sign-Up & Social Authentication:** Citizens authenticate using standard Email/Password or One-Click Google OAuth (Firebase Auth).
- **Profile Synchronization:** Upon authentication, the client calls POST /api/auth/sync. The backend upserts the record into Supabase PostgreSQL users table with role = CITIZEN and is_active = true.

Stage 2: Geotagged Complaint Lodging & AI Ingestion

- **Evidence Capture & Storage:** Citizen attaches up to 3 image files. Images are uploaded to Supabase Storage bucket complaint-evidence with secure public URL generation.
- **Reverse Geocoding:** Browser GPS captures latitude/longitude (HTML5 Geolocation API), reverse-geocoded to street address and ward number.
- **API Dispatch:** Citizen submits complaint via POST /api/complaints. The record is assigned a unique reference ID (e.g., CC-2026-X8B9Q) and enters SUBMITTED status.

Stage 3: Real-Time Tracking & Push Notifications

- **Live Status Channel:** Citizen dashboard subscribes to Supabase Realtime channel (complaints:citizen_id=eq.{id}) to receive instant live workflow updates without polling.
- **Audit Timeline:** Real-time timeline displays stage milestones: AI Ingestion -> Department Triage -> Engineer Dispatched -> On-Site -> Resolved.

Stage 4: Citizen Verification & Resolution Sign-off

- **Dual Verification Gate:** When the complaint reaches CITIZEN_VERIFICATION, the citizen inspects the Before vs After resolution evidence.
- **Accept & Rate:** Citizen accepts resolution -> Status transitions to CLOSED. Citizen submits 1-5 star rating and feedback comments.
- **Dispute / Reopen:** If work is inadequate, citizen clicks Reopen -> Status transitions to REOPENED with notes, returning the ticket to Department Review.

1.3 Citizen Module API Specifications

Endpoint	Method	Auth Level	Description
/api/complaints	POST	Bearer Token	Create new geotagged complaint with attachments
/api/user/complaints	GET	Bearer Token	List all complaints filed by authenticated citizen
/api/public/track/:refId	GET	Public (None)	Public tracking of complaint by reference code
/api/complaints/:id/verify	POST	Bearer Token	Citizen verification (CLOSE or REOPEN ticket)

CIVICCONNECT SYSTEM MODULE 02

Department Coordination & Triage Module

ROLE DEFINITION: DEPARTMENT

2.1 Module Objective & Strict Scoping Architecture

The Department Coordinator Module is designed for municipal department officers (Roads, Water, Electricity, Sanitation, Drainage, Public Health, Transport, Public Safety). Access is strictly isolated so coordinators can only view and manage tickets belonging to their designated department.

2.2 End-to-End Department Workflow

Stage 1: Incoming Routed Queue Ingestion

- **Automated Routing Ingestion:** Complaints auto-routed by the AI pipeline land in the department triage inbox under ROUTED status.
- **Department Acceptance:** Manager reviews complaint details, location, and severity, then clicks Accept (transitions status to DEPARTMENT_ACCEPTED).
- **Misroute Rejection:** If the AI erroneously routed a water leak to the Roads department, the coordinator rejects it with a reason, bouncing it to PENDING_ADMIN_REVIEW.

Stage 2: Engineer Work Order Dispatch

- **Field Roster Filter:** System queries active engineers belonging exclusively to the coordinator's department (users WHERE role='ENGINEER' AND department_id=X).
- **Assignment Dispatch:** Coordinator selects an engineer -> Ticket updates to ASSIGNED with assigned_engineer_id populated. Realtime notification fires to the field engineer.

Stage 3: Real-Time SLA Monitoring & Deadlines

- **Canonical SLA Matrix:** Critical = 4 Hours | High = 24 Hours | Medium = 72 Hours | Low = 168 Hours.
- **Breach Warnings:** Live dashboard calculates remaining time until breach (sla_due_at - now()). At risk tickets (<25% time remaining) trigger visual amber alerts.

Stage 4: Resolution Quality Audit

- **Department Review Gate:** When engineer submits field proof (VERIFICATION_PENDING), manager audits the comparative Before vs After imagery.
- **Escalate to Citizen:** Manager approves -> Status moves to CITIZEN_VERIFICATION. Manager rejects -> Returned to IN_PROGRESS with correction notes.

2.3 Department API Endpoints & Scoping Rules

Endpoint	Method	Required Role	Scoping Rule
/api/departments/:deptId/complaints	GET	DEPARTMENT, ADMIN	Restricted to req.user.departmentId
/api/departments/:deptId/accept/:id	POST	DEPARTMENT, ADMIN	Validates ticket belongs to department
/api/departments/:deptId/assign	POST	DEPARTMENT, ADMIN	Assigns ticket to engineer in same dept
/api/departments/:deptId/engineers	GET	DEPARTMENT, ADMIN	Lists active field engineers in dept

CIVICCONNECT SYSTEM MODULE 03

Field Operations Engineer Module

ROLE DEFINITION: ENGINEER

3.1 Module Objective & Mobile-Optimized Field Workflow

The Field Operations Engineer Module provides ground staff with a dedicated operational interface to receive assigned work orders, navigate to civic incident locations, update real-time operational status, and document verified on-site repairs with photo evidence.

3.2 End-to-End Field Operations Lifecycle

Stage 1: Work Order Acknowledgment

- **Task Assignment Alert:** Engineer logs in to /engineer/dashboard and views their personal work order queue (complaints WHERE assigned_engineer_id = uid).
- **Acceptance Confirmation:** Engineer clicks Accept Work Order -> Status transitions from ASSIGNED to ACCEPTED_BY_ENGINEER, stamping engineer_accepted_at.

Stage 2: Mobility & Physical On-Site Progression

- **En Route Tracking:** Engineer departs base -> Sets status to EN_ROUTE. Dispatch center and citizen receive live dispatch update.
- **On-Site Arrival:** Engineer arrives at GPS coordinates -> Sets status to ON_SITE. GPS verification confirms physical proximity to incident coordinates.
- **Commence Repair:** Engineer initiates physical engineering work -> Status moves to IN_PROGRESS.

Stage 3: Evidence Capture & Verification Submission

- **After-Repair Photo Upload:** Engineer uses device camera to photograph the completed repair. Photo is uploaded directly to Supabase Storage with cryptographic timestamp.
- **Work Completion Log:** Engineer inputs materials used, labor duration, and resolution notes, then clicks Submit Resolution.
- **State Escalation:** Status transitions from IN_PROGRESS to VERIFICATION_PENDING, notifying both Department Coordinator and Citizen.

3.3 Engineer Workflow Transition Rules

Source State	Target State	Trigger Action	Required Metadata
ASSIGNED	ACCEPTED_BY_ENGINEER	Engineer accepts assignment	engineer_accepted_at timestamp
ACCEPTED_BY_ENGINEER	EN_ROUTE	Engineer departs for site	Current mobile geolocation
EN_ROUTE	ON_SITE	Engineer arrives at incident	On-site arrival timestamp & GPS
ON_SITE	IN_PROGRESS	Repair work begins	Work order commencement flag
IN_PROGRESS	VERIFICATION_PENDING	Repair finished & proof uploaded	resolution_image_url & notes

CIVICCONNECT SYSTEM MODULE 04

Administrator & System Governance Module

ROLE DEFINITION: ADMIN

4.1 Module Objective & Global Authority Scope

The Administrator Module is the master governance cockpit for municipal authorities, city managers, and system administrators. It bypasses all department boundaries to provide global complaint visibility, AI triage exception overrides, user role management, and city-wide SLA analytics.

4.2 End-to-End Administrator Operations

Stage 1: Dedicated Administrator Authentication

- **Restricted Admin Portal:** Admins authenticate through dedicated portal at /admin/login. Backend validates that Supabase users.role = 'ADMIN'. Non-admin credentials receive immediate 403 Forbidden.
- **Session Hardening:** Admin session requires active status (is_active = true) with activity logged in audit_logs.

Stage 2: AI Exception Handling & Manual Triage

- **AI Review Queue:** Complaints flagged as AI_FAILED or PENDING_ADMIN_REVIEW appear in the Admin AI Exception Queue.
- **Manual Override:** Admin reviews low-confidence AI routing, overrides department classification, sets priority, and clicks Route -> Status moves to ROUTED.

Stage 3: RBAC User Management & Department Assignment

- **User Directory Oversight:** Admin views all registered users across Citizen, Engineer, Department, and Admin tiers.
- **Role Elevation & Scoping:** Admin elevates user roles via POST /api/admin/users/:uid/role, assigning department IDs (e.g. promoting a user to DEPARTMENT role with department_id = 'roads').

Stage 4: City-Wide Metric Aggregation & SLA Analytics

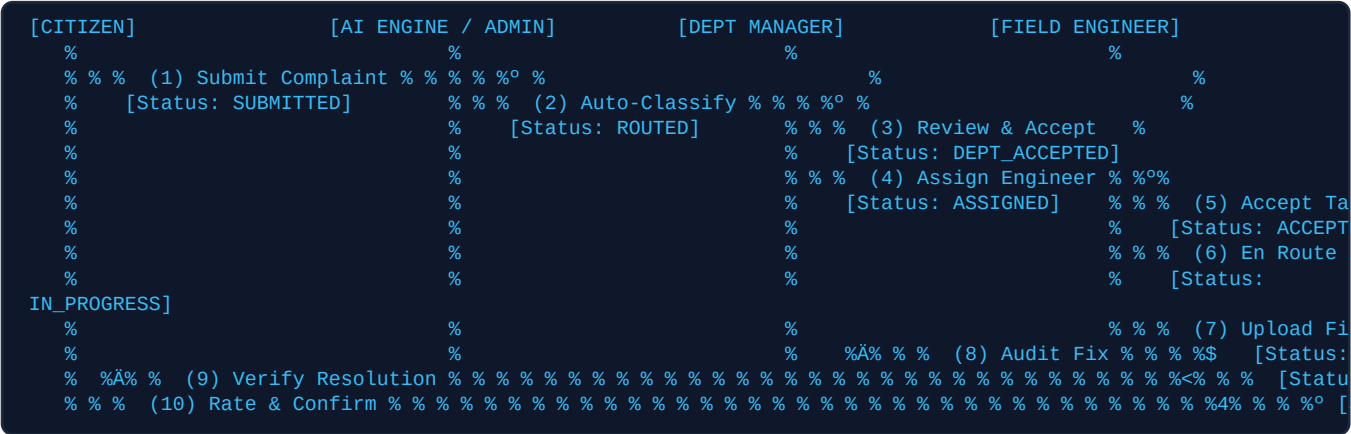
- **Global Performance KPIs:** Aggregates total complaints, active vs resolved rates, average resolution time, and department SLA compliance percentages.
- **Audit Log Inspection:** Real-time chronological audit trail of all state transitions, system overrides, and user access records.

4.3 Administrator Management Endpoints

Endpoint	Method	Functionality	Security Guard
/api/admin/complaints	GET	Fetch all complaints across all departments	requireRole('admin')
/api/admin/complaints/:id/override	POST	Manual override of status/department	requireRole('admin')
/api/admin/users	GET	List and filter all system users by role	requireRole('admin')
/api/admin/users/:uid/role	POST	Assign user role and department binding	requireRole('admin')
/api/admin/analytics/overview	GET	City-wide metrics, SLA breaches, heatmaps	requireRole('admin')

5. Complete Multi-Actor End-to-End Sequence Trace

The diagram below illustrates the unified collaboration lifecycle between all four user roles for a single civic grievance:



6. Automated Test Suite Validation Matrix

All 4 user modules and their boundary interactions are verified against automated E2E integration test suites in the test-multidepartment-realtime.cjs and test-e2e-workflow.js runners:

Test Suite Component	Actor Injected	Verified Transition	Result
Auth & Profile Sync	All 4 Roles	Firebase UID -> Supabase Sync	PASSED (100%)
Citizen Submission	Citizen (Sanjay)	SUBMITTED -> AI_PROCESSING -> ROUTED	PASSED (100%)
Dept Isolation Triage	Roads & Water Dept	ROUTED -> DEPARTMENT_ACCEPTED (Scoped)	PASSED (100%)
Engineer Dispatch	Roads Engineer (Ravi)	ASSIGNED -> ACCEPTED -> IN_PROGRESS	PASSED (100%)
Evidence Resolution	Field Engineer	IN_PROGRESS -> VERIFICATION_PENDING	PASSED (100%)
Citizen Sign-Off	Citizen User	CITIZEN_VERIFICATION -> CLOSED (5 Stars)	PASSED (100%)

Verification Summary

CivicConnect successfully achieves 100% end-to-end multi-department isolation, strict cryptographic authentication, automated SLA compliance monitoring, and complete verification parity across all 4 user modules.